

Bi-directional imagined hand movement classification using low cost EEG-based BCI

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Abstract— The notion of developing thought controlled devices (games, robots, cars etc.) is becoming increasingly popular with the introduction of low cost commercial headsets that record neuroelectric activity and the extensive research in the area of Brain Computer Interfaces (BCIs). In this paper, we study the feasibility of using a commercial low cost EEG amplifier which has only limited number of electrodes, to develop a motor control BCI system. The objective is to extract brain activity responsible for direction specific imagined and executed motor activity, which can be used to identify the motor task performed by the user using the simultaneously recorded EEG. An experiment is conducted to engage the user in bi-directional horizontal movement execution and imagination of the dominant hand. The analysis includes investigation of the time-frequency bins of the recorded EEG that provides maximum discrimination of directional movement. Further, the features are extracted using Filter Bank Common Spatial Pattern (FBCSP), followed by Fisher Linear Discriminant (FLD) for classification. The classification performance at various time instants of each trial are considered; and a control strategy was introduced at the classifier output to enhance performance. The performance in terms of average classification accuracy over five subjects is obtained as 81.3 % (movement execution) and 82.4 % (movement imagination). The results indicate the applicability of this EEG-BCI system to provide directional motor control to an interfaced device such as a robotic arm or a game element.

Keywords—Brain Computer Interfaces, EMOTIV EPOC, motor control BCI

I. INTRODUCTION

Brain Computer Interfaces (BCIs) translate the scalp-recorded electrical signals that represents brain activity into control outputs that allow user to communicate with external environment without using peripheral nerves or muscles [1]. The ultimate goal of BCI is to develop assistive devices that allow communication and control for people with neuromuscular disorders [1, 2]. BCI research focuses on wide range of human activity such as locomotion, cognition, perception [3, 4] etc. Motor control BCI systems aim to identify the neural correlates underlying motor tasks including intention, preparation, execution and imagination of a movement task. Electroencephalography (EEG) is the most commonly used data acquisition system in BCI research and EEG-BCI systems explore the neuro-electrical signal features underlying human activities.

The limited number of control signals that can be derived from EEG is one of the major challenges in developing motor control BCI system [5]. Many researchers have developed EEG-BCI systems using various sophisticated signal processing and machine learning tools to classify specific limb movements [5-8]. Few studies [9-11] have investigated movement parameter (speed, force, direction etc.) information present in EEG, and have performed continuous reconstruction of movement paths. However, identifying the neural correlates of multi-directional imagined movement is a less explored area of EEG-BCI research. Such tasks engage brain in high cognition levels to simultaneously perform spatial coordination and kinesthetic imagery. The studies in [12, 13] have reported the reconstruction of multi directional imagined movement in a 3D space. This type of fine motor control BCI can highly impact neurorehabilitation applications.

In this study, we investigated the feasibility of developing an EEG-BCI system to classify bi-directional hand movement. Such a system could be interfaced to an assistive device, a robotic arm or entertainment/gaming application to achieve directional motor control solely using user's thoughts (brain activity). Further, in the present study, we investigated the applicability of a commercial and low cost EEG amplifier, namely EMOTIV EPOC, to develop a motor control BCI system. The challenges here involved the lack of electrodes placed in the motor cortex and the limited number of electrodes available for data acquisition. However, this neuroheadset has been popularly used for emotion recognition, attention and cognition enhancement and multimedia applications [14-16]. Further, the studies [17, 18] have reported motor control BCI that can classify bilateral hand movement imagination from signals recorded using this amplifier. Both these works mention the limitations of EMOTIV EPOC to provide signal quality comparable to professional research grade EEG amplifiers, which has resulted in error sensitivity of the BCI developed. However, with the introduction of post-processing tools [17] and advanced classification tools [18], classification accuracies comparable to the more sophisticated amplifiers have been achieved.

Motivated by this factor, in this study we used the neural signals recorded by EMOTIV EPOC amplifier to develop motor control BCI that investigates center-out motor tasks. The motor tasks studied were bi-directional hand movement execution and imagination using the user's dominant hand. The proposed BCI incorporated signal processing tools to enhance

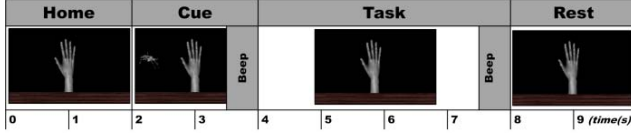


Fig. 1. Experiment timeline for a single-trial.

signal quality and to identify the discriminative features between the two movement directions. The classified output was then translated to an output control command corresponding to the targeted movement direction. The interface was developed in format of a BUG-KILL game, in which the user hits a bug appearing on either left hand or right hand side of the screen, by imagining center-out hand movement towards its location. The proposed system used the simultaneously recorded brain signal to identify the movement direction. This paper reports the initial phase of this system development which investigated the feasibility of using the EMOTIV EEG signals for motor-based neural feature identification.

The paper is organized as follows: The details of the experiment conducted is given in Section II. Section III explains in detail the processing tools used in the study. Section IV discusses the major findings and their significance, along with the limitations of current work and the scope of future work. This is followed by Section V that concludes the paper.

II. BCI EXPERIMENT

The goal of the study was to design BCI system to classify bi-directional hand movement from simultaneously recorded EEG signal. An experiment protocol incorporating the tasks, timelines, visual cues etc., was designed to guide the user through brain data acquisition. Data thus acquired were used for system development and validation. The details of the experiment are given in the following sub-sections.

A. Subjects and Equipment

Five healthy subjects (4 males, 1 female; all right-handed;

aged 25-30) participated in the experiment and data acquisition. (The subjects were given IDs, namely, RVW, SJS, SKG, AAB, RRV). EMOTIV EPOC neuroheadset was used to record the EEG activity. The signal from all 14 electrodes (placed according to 10-20 international system of EEG electrode placement) was acquired. In order to acquire signals from pre-motor/motor areas, the headset was placed slightly backwards. The signal was recorded at a sampling rate of 128 Hz and band pass filtered at 0.2-45 Hz.

B. Experiment: Interface and Protocol

The motor tasks investigated in the study were imagination and execution of center-out movements in horizontal plane towards right-left directions. The experiment design included four experiment sessions of 30 trials each. Each session was separated by a time interval of 5 minutes. The subjects were instructed to execute hand movement using their dominant hand during experiment sessions 1 and 2, and to imagine similar movement during experiment sessions 3 and 4.

During the experiment, the subject was seated in front of a monitor displaying the experiment interface. In order to engage the user in the experiment, the visual interface was developed in format of BUG-KILL game. The user rested his hand comfortably on the table and was instructed to maintain eyes fixated at center of the screen throughout each session. The interface presented image of a hand at the center of screen, to resemble the user's posture and served as the eye-fixation point. In the interface, a bug appeared in the right hand or left hand side of the displayed hand. The user's objective was to either move his hand (during sessions 1 and 2) or imagine its movement (during sessions 3 and 4) to the bug's position (so as to hit it). The movement was done in horizontal plane along the surface of the table.

The experiment timeline, tasks and visual cues are indicated in Fig.1. Each trial started with a home screen (hand displayed at center). After two seconds, the target was presented indicated by position of bug. At the end of 2 seconds, a beep sound was played and the bug was removed. This served as the cue for the user to start movement task. This task segment was four seconds long, at the end of which another beep was played, that informed the user to return hand to the center position. A rest segment of 2 seconds followed after which this cycle repeats.

All the participants engaged in four sessions of the experiment. Each session consisted of 30 trials, 15 right and 15 left targets distributed in random.

III. METHODOLOGY

The objective of this study was to examine the feasibility of using commercial devices, such as, EMOTIV EPOC neuroheadset to identify and classify the movement tasks. This section provides the signal analysis tools used in the proposed BCI system. These tools function to convert brain activity to an external motor control command.

The functional units in the proposed BCI are indicated in Fig. 2. The input unit was the data acquisition unit, which recorded EEG, as the user performed required task. The EEG data was then passed on to the signal analysis block which

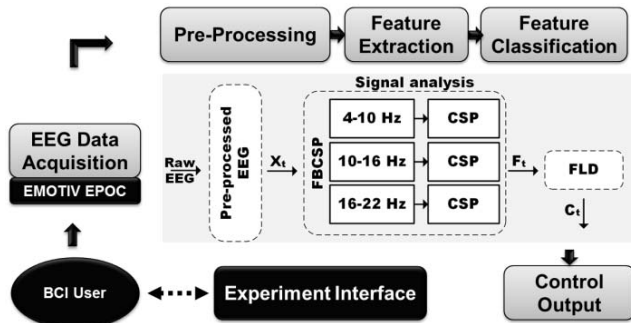


Fig. 2. BCI functional units.

consisted of pre-processing, features extraction and classification units. The pre-processing steps involved baseline correction and spatial smoothening using Laplacian filters [19]. A time-frequency analysis of the pre-processed EEG was performed, in order to identify its information content. Further, the pre-processed EEG was used to extract discriminative features corresponding to the motor tasks performed, and classified to evaluate BCI performance. The description of the signal processing units are given in the following sub-sections.

A. Discriminative time-frequency bins

Sensorimotor rhythm based features are widely used to develop motor control BCI using EEG. The power suppression and enhancement – Event Related (De) Synchronization (ERD/ERS) – of mu (8-12 Hz) and beta (18-26 Hz) bands from the sensorimotor areas of the brain are distinct indicators of motor activity [20]. Such ERD/ERS activity are however specific to the limb/body part engaging in the motor task. However, in the current study, we investigated bi-directional movement of a single hand, and hence, time and frequency specific correlates of this movement task had to be investigated.

In this study, the metric Signal-to-Noise Ratio (SNR) was used to quantify the discrimination between two directions of movement provided by time-frequency (TF) localized EEG. The metric has been used in [10, 11] to analyze the TF activations in multi-directional movements. The power spectral density of each trial, PSD_n was determined including frequencies from 4-20 Hz and 2-10s (from Fig.1) of the trial. PSD was calculated using 250ms time window with an overlap of 62.5ms; and non-overlapping spectral bands of 0.5 Hz frequency bandwidths.

The SNR metric consisted of three components namely, the signal power, V_s , noise power, V_a and bias correction term, V_b . The signal power was calculated by the variance between trial-averaged PSD for each movement class. The noise power was obtained as the trial-by-trial fluctuations in PSD, that is, the variance between PSD of all the trials. The bias term was calculated as the trial-by-trial fluctuations within class-specific PSD. With these components, SNR can be defined for every TF bin as,

$$SNR = \frac{(V_s - V_b)}{V_n} \quad (1)$$

The SNR for each of the sensor location was thus calculated and the average SNR across all the sensors was used to illustrate discriminative TF bins. Further, the TF pattern was smoothened using a sliding window mean filter, for better visualization.

B. Feature extraction

A popular method to extract EEG signal features that can discriminate motor activity is Common Spatial Pattern (CSP) [21]. This algorithm optimally discriminates binary class of data by the simultaneous diagonalization of two covariance matrix. Many variants of CSP have been introduced such as Filter Bank CSP (FBCSP) that allows spectral localization of CSP by extracting CSP features from pre-defined spectral bands [22]. In this study, we used FBCSP to discriminate bi-

directional hand movement. The three spectral bands used are shown in Fig. 2.

The time segment extracted from pre-processed EEG for further processing was 2-10 s as indicated in Fig. 1. This data was divided into 8 time segments of one second each. The data in each segment, t is denoted by $X_t (\in \mathbb{R}^{C \times S})$ in Fig. 2, where C and S indicates the number of channels and samples per trial respectively. For feature extraction using FBCSP, the signal was spectral filtered as indicated in Fig. 2, using Chebyshev Type-II filter. The subband data was spatially filtered using CSP. The CSP filtered EEG from a single trial data can be denoted as,

$$Z_t = W_t \cdot X_t \quad (2)$$

For further analysis, the first and last m rows of Z_t , which provided maximum discrimination in variance between two classes of data. The log-normalized variances of these $2m$ rows were the features obtained. In Fig. 2 this is denoted by F_t .

C. Classification and control output

Feature extraction was followed by classification stage in which, the motor activity corresponding to the extracted EEG features was identified. Fisher Linear Discriminant (FLD) [23] classifier was used in this study. FLD calculated a discriminant function that maximized the ratio of between-class scatter to within-class scatter. The classifier output was set as a binary value, 1 for a correct classification and 0 otherwise. In Fig. 2, this binary output obtained from each time segment is denoted by C_t .

In this study, subject-specific variations of the EEG temporal activity were considered. It was observed that the t in which each subject achieved his maximum BCI performance was a variable and this affected the average classification performance. Hence taking this variation into account, a control strategy was introduced at the output stage of the BCI. The methods to obtain the classification output, $\hat{C}$ is explained below:

- Segmented-Approach (Seg): The classification accuracy for each time segment was separately calculated. The average and the maximum accuracy achieved across the subjects were reported. The subject-specific time segments that provided highest accuracy were also noted.
- Accumulated-Approach (Accu): For each trial, classifier output was accumulated as $\{C_1, \dots, C_t, \dots, C_K\}$. The control strategy determined the classifier output corresponding to each trial as,

$$\hat{C} = \begin{cases} 1, & \sum_t C_t \geq K/2 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Here K is the number of elements in the accumulated classifier output and ($=8$) as all the time segments were considered.

- Selectively accumulated-Approach (selAccu): The above method in (3) can further be optimized to use only subject-specific time segments in which high classification (in terms of accuracy) was achieved. In this study, the results of the segmented approach were used to select the subject-

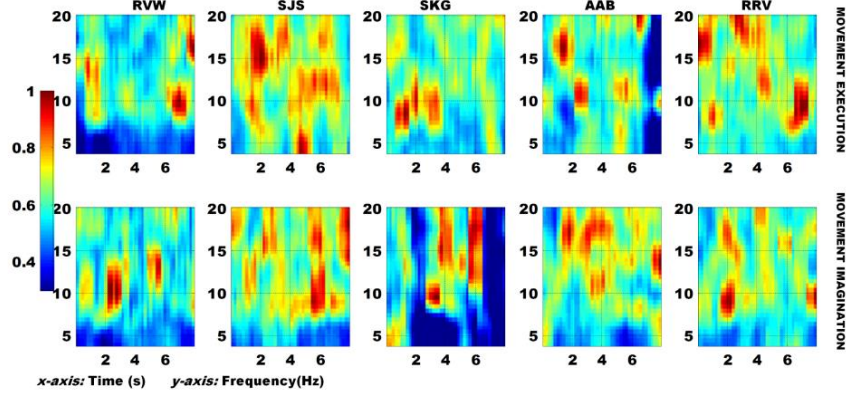


Fig. 3. Discriminative TF patterns using SNR. The columns indicate the subjects and rows indicate motor tasks.

specific time segments. The number of segments selected was denoted by k and the accumulated classifier output can be denoted by $\{C_1, \dots, C_t, \dots, C_k\}$. The classifier output determined as,

$$\hat{C} = \begin{cases} 1, & \sum_t C_t \geq k/2 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

D. Data analysis

The data from sessions 1 and 2 were concatenated to obtain movement execution dataset and the data from sessions 3 and 4 to obtain movement imagination dataset. For the validation of the proposed BCI system, 5×5 cross-validation was performed. The training samples in each cross-validation fold were used to determine the system parameters such as, CSP filter W_t (for all 3 subbands) and the classifier model. These parameters were applied to evaluate the class label for the test data samples.

IV. RESULTS AND DISCUSSION

The results of the proposed system validation and the findings of the analyses are reported in this Section.

A. Time-frequency SNR patterns

The TF distribution of SNR from each sensor location can

be calculated using (1). The sensor-averaged and normalized SNR for each of the subject during movement execution and imagination task is displayed in Fig. 3. The warm regions represent high-SNR values, indicating the TF bins that provided discriminative information regarding direction of hand movement. The time segment from 2-6s indicates the movement task period, prior to which is the preparation time and following it is the rest period.

In all the subjects, higher activations are observed in mu and beta spectral regions. From the SNR patterns of user RVW (column 1 of Fig. 3), it can be observed that an activation occurs at the beginning and towards end of movement, which could correspond to ERD/ERS. The patterns for users SJS and RRV lack localization, but still show highest SNR values towards pre/post movement time segments. The TF patterns obtained using SNR clearly indicate the presence of discriminative information regarding movement direction (imagined or actual) in the EEG signals. However, the lack of correlation between imagined and executed motor patterns, and the variations between subjects, can also be observed from the figure.

B. Classification performance

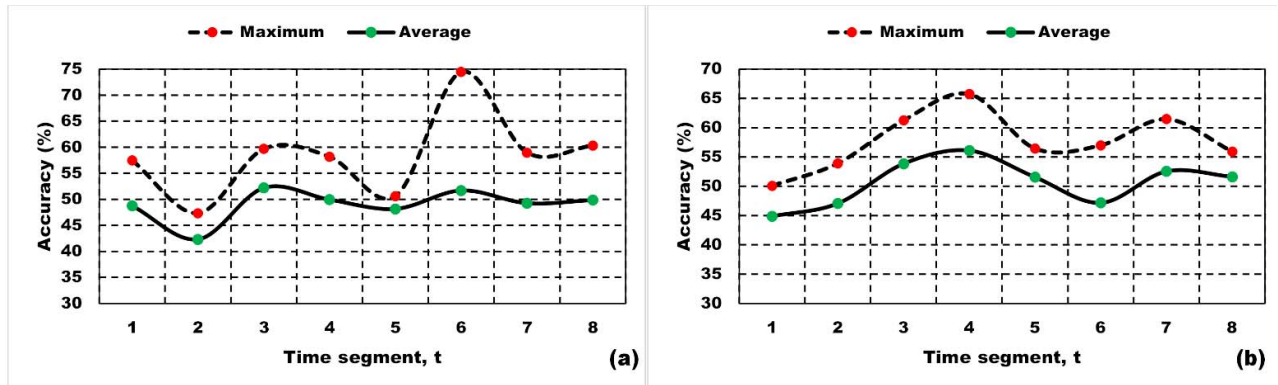


Fig. 4. Performance in terms of classification accuracy using Seg-Approach: (a) Motor execution (b) Motor imagination.

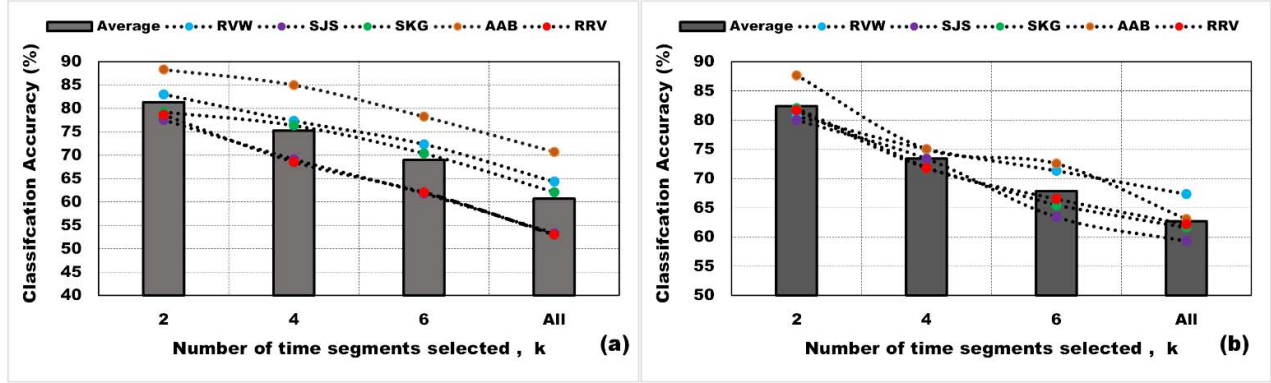


Fig. 5. Classification performance for selected time segments: (a) Movement execution (b) Movement imagination

The results of classification approach along with the control strategy adopted in this study is reported in this section. The performance is measured in terms of classification accuracy and its average over subjects. The results of the Seg-Approach is displayed in Fig. 4. It can be observed that the average classification accuracy is below chance level in some time segments. This is clearly due to the difference in time segment of activation in various subjects and these results demonstrate the need for control strategy proposed in this paper. Further, the dotted plot in Fig. 4 shows the maximum accuracy among 5 subjects achieved in each of the time segment, t . Averaging this over all the time segments, 58.4 % (execution) and 57.7 % (imagination) classification accuracies are obtained. These results are comparable to the values reported in classification of right hand versus left hand imagery in [17, 18] using EMOTIV EPOC amplifier.

The control strategy mentioned in section III-C, namely Accu and selAccu approaches are further used for classification and the results are as follows. In Accu-approach, $K = 8$ and selAccu-approach, we set the number of time segments chosen as $k = \{2, 4, 6\}$, to study their effect on performance. The results of both methods when used to classify bi-directional hand movement execution and imagination are plotted in Fig. 5. It should be noted that Accu-approach is same as that of selAccu-approach with $k = 8$. The steady increase in classification accuracy achieved by using selected time segments can be observed from the results. The performance of each of the participants are also indicated in the figure.

The results clearly indicate the advantages of using the control strategy considering the classification results at different time segments of a single trial. This strategy effectively replaces the need for temporal localization of the feature extraction algorithm. Average accuracies of 81.3 % (execution) and 82.4 % (imagination) are achieved using the proposed method. The results indicate the usability of this strategy to online BCI applications for bi-directional motor control commands.

C. Limitations and scope for future work

The selection of subject-specific time segments and spectral regions that provides optimally discriminative features is one of the challenges in developing EEG-BCI system. In this study, an important component in the proposed classification and control output generation, is the selection of informative time segments. The results reported here are based on the time segments that provided high classification accuracy in Seg-Approach. However, optimization algorithms that can select active areas in time domain can be explored to replace this selection. Further, these time segments can also be calculated using the TF maps mentioned in III-A. The results reported in Section IV-A, clearly indicate the presence of discriminative time segments.

As previously mentioned, this is the initial phase of our study to develop BCI system using commercial EEG amplifier to provide bi-direction motor control to an interfaced device. Our results indicate that such a system is feasible using the proposed control strategies. An accuracy over 80% is achieved in both motor tasks in the present study. Future study will focus on implementing closed loop BCI, in which the output, $\hat{C}$ for every trial will be provided as feedback to the user. The effect of neurofeedback to increase user's control over directional movement will also be studied. Thus the BUG-KILL game interface in real-time scenario with feedback, is sure to motivate and engage the user in brain-training.

V. CONCLUSION

The study investigated the applicability of using a commercial EEG amplifier to develop a motor control BCI system. Bi-directional movement imagination and execution of dominant hand were studied using the proposed EEG-BCI system. To the best of our knowledge, a study with these objectives has not yet been reported in the literature. The EEG data for validation was acquired as the subject engaged in a visual interface designed as per the proposed experiment protocol. The data was analyzed to identify the TF bins with maximum direction specific information. Further feature extraction and classification was performed along with the proposed control strategy to enhance the classification performance. The results achieved indicate the usability of this

interface for gaming/entertainment applications. The direction control that can be achieved from EEG-BCI thus takes us one step closer to the development of thought-controlled devices.

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